Introductory AI BootCamp

Program Overview

Artificial Intelligence, Machine Learning and Deep Learning overview – Python ecosystem for AI (NumPy, Pandas, Matplotlib) – Mathematical foundations (linear algebra, probability, optimization basics) – Data understanding and preprocessing (cleaning, feature engineering, normalization, train-test split) – Feature extraction from images, text and signals – Fundamentals of machine learning (model training, bias–variance tradeoff, overfitting and underfitting, evaluation metrics) – Supervised learning algorithms (Linear Regression, Logistic Regression, k-NN, Decision Trees, Random Forest, Support Vector Machines) – Unsupervised learning methods (clustering with K-Means, dimensionality reduction using PCA) – Introduction to neural networks (perceptron, multilayer networks, activation functions) – Training deep neural networks (forward propagation, backpropagation, gradient descent, regularization) – Deep learning frameworks and implementation using TensorFlow/Keras – Convolutional Neural Networks (CNN) for image data – Sequence models including Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks.

Learning Objectives

- Introduce the fundamental concepts of AI, ML, and DL.
- Develop a foundation in the mathematical and computational principles underlying ML.
- Familiarize participants with the Python ecosystem for AI including NumPy, Pandas, Matplotlib and Scikit-learn.
- Enable participants to understand data preprocessing, feature extraction, and model evaluation techniques.
- Provide experience implementing supervised and unsupervised ML algorithms.
- Introduce neural networks and deep learning model architectures.
- Demonstrate deep learning frameworks such as TensorFlow and Keras.
- Expose participants to CNNs and sequence models such as RNN and LSTM.
- Encourage project development and research-oriented thinking in AI.

Learning Outcomes

- Understand the core workflow of machine learning and deep learning systems.
- Apply Python libraries to implement machine learning models.
- Perform data preprocessing, feature extraction, and exploratory data analysis.
- Train and evaluate supervised and unsupervised machine learning models.
- Build neural network models using deep learning frameworks.
- Implement CNN models for image data and RNN/LSTM models for sequential data.
- Develop small AI projects integrating theory and coding.
- Gain confidence to pursue further research or careers in AI and data science.

BootCamp Sessions (18 Sessions)

Session	Topic	Activity
1	Introduction to Artificial Intelligence, Machine Learning and Deep Learning	Discussion of real-world AI applications and Python environment setup
2	Python Ecosystem for AI (NumPy, Pandas, Matplotlib)	Hands-on data manipulation and visualization
3	Mathematical Foundations I (Vectors and Matrices)	Implement vector and matrix operations using NumPy
4	Mathematical Foundations II (Probability and Optimization)	Probability simulations and distribution visualization
5	Data Understanding and Preprocessing	Dataset cleaning, missing value handling, normalization
6	Feature Extraction from Images, Text and Signals	Extract basic features from sample datasets
7	Machine Learning Fundamentals	Bias–variance visualization and evaluation metrics exploration
8	Linear Regression and Logistic Regression	Build prediction and classification models using Scikit-learn
9	k-NN and Decision Trees	Implement classification models and analyze results
10	Random Forest and Support Vector Machines	Train ensemble models and compare performance
11	Unsupervised Learning: Clustering	Perform clustering using K-Means
12	Dimensionality Reduction using PCA	Reduce dimensions and visualize clusters
13	Introduction to Neural Networks	Build a simple perceptron / MLP model
14	Training Deep Neural Networks	Implement gradient descent and learning visualization
15	Deep Learning Frameworks (TensorFlow / Keras)	Build and train neural networks using Keras
16	Convolutional Neural Networks (CNN)	Image classification experiment
17	Sequence Models: RNN	Implement simple RNN for sequential data
18	Sequence Models: LSTM and Project Discussion	Build sequence prediction model and discuss project ideas